An Internship Report

Of

Siemens Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030153

Name of Student: SAGAR RANA

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, SAGAR RANA, do hereby declare that the internship report titled “Siemens Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-
Student Name: SAGAR RANA
Roll No.: 2410030153

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and Siemens for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-
Student Name: SAGAR RANA
Roll No.: 2410030153

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शिक्षा मंत्रालय
MINISTRY OF EDUCATION
सत्यमेव जयते

अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills®
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

SAGAR RANA

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai	Dr. Buddha Chandrasekhar	Shubhajit Jagadev
Head - Strategy and Operations	Chief Coordinating Officer (CCO)	Chief Executive Officer (CEO)
Siemens Industry Software (India) Pvt. Ltd	AICTE, Ministry of Education	EduSkills

Certificate ID : 43cab78d02bbc5d5051e

Student ID: STU6836f1b4b1fdc1748431284

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

INTERNSHIP OFFER LETTER

EduSkills®
Nation Building Through Skills

Ref No: C17Y2026M071502171

Internship Offer Letter

Student Name	: SAGAR RANA
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU6836f1b4b1fdc1748431284

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

Scan to Verify

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “Siemens Data Science Master”, supported by Siemens Industry Software (India) Pvt. Ltd. The internship was carried out from June 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong practical foundation in data science and applied machine learning, progressing from identifying business problems suitable for AI, through data engineering and model building, to enterprise-level deployment and administration of a data science platform.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on modules covering applications and use cases, data engineering, machine learning, data preparation, and platform administration, culminating in a Final Credential Validation / Internship Grade Point Assessment. On successful completion of all the requirements, a Virtual Internship Certificate with an “Outstanding” grade was awarded by EduSkills Academy under the patronage of AICTE and the National Internship Portal.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Data Science Master track, EduSkills collaborates with Siemens Industry Software (India) Pvt. Ltd. to design an industry-aligned curriculum that helps engineering students build practical data science and machine learning capability using enterprise-grade tools such as the RapidMiner AI Hub.

4.3 Problem Statement

Although a large number of engineering students learn the theoretical foundations of statistics, data handling, and machine learning during their academic coursework, very few get structured, hands-on exposure to running a complete, enterprise-style data science workflow – from identifying a business problem, through data access and transformation, to building, evaluating, deploying, and monitoring machine learning models on a managed platform. The problem addressed by this internship was to practically learn and apply this end-to-end data science workflow using an industry-standard platform, and to understand how such a platform is administered and operated at an enterprise level.

4.4 Project Objectives

- To learn to identify business problems suitable for AI and machine learning solutions.
- To learn to access, load, transform, and calculate data as part of a structured data engineering workflow.

- To build foundational machine learning skills, including model selection and evaluation.
- To master advanced data handling, cleaning, and automation techniques.
- To learn advanced machine learning techniques for improved model performance.
- To learn to deploy and monitor machine learning models in a production-like setting.
- To configure and manage an enterprise AI platform (RapidMiner AI Hub).
- To understand enterprise-level platform administration and operations.
- To successfully clear the Final Credential Validation / Internship Grade Point Assessment.

4.5 Scope of the Project

The scope of this internship spans the complete data science lifecycle – starting from identifying business use cases for AI and machine learning, moving through data engineering (accessing, loading, transforming, and calculating data), and into building and refining machine learning models using both foundational and advanced techniques. It further covers deploying and monitoring models in applied, use-case-driven scenarios, and configuring and administering an enterprise AI platform. The internship was limited to a virtual, simulation/lab-based learning environment provided by EduSkills Academy in association with Siemens, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Data Science Platform: RapidMiner AI Hub
- Core Areas: Data Engineering, Data Preparation, Machine Learning, Model Deployment and Monitoring
- Techniques: Business use-case identification for AI/ML, data access and transformation, foundational and advanced machine learning, automation of data handling
- Platform Operations: Configuration and management of the RapidMiner AI Hub, enterprise-level platform administration
- Learning Platform: AICTE – EduSkills Virtual Internship Program modules and weekly assessments

4.7 System Architecture

The general workflow followed during the internship for the Siemens Data Science Master track can be summarized in the following layers:

- Use-Case Layer: Business problems are analysed and mapped to suitable AI and machine learning approaches.
- Data Engineering Layer: Data is accessed, loaded, transformed, and calculated to prepare it for modelling.

- Data Preparation Layer: Advanced data handling and automation techniques refine and structure the data for modelling.
- Modelling Layer: Foundational and advanced machine learning techniques are applied to build predictive models.
- Deployment & Monitoring Layer: Trained models are deployed and continuously monitored for performance.
- Platform Administration Layer: The RapidMiner AI Hub is configured and managed to support enterprise-level platform operations.
- Assessment Layer: Weekly assessments, project documentation, and a Final Credential Validation / Internship Grade Point Assessment track progress and final grading.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on modules, and weekly assessments, culminating in a Final Credential Validation, as summarized below.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
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Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There was no stipend associated with this internship.

Each week concluded with assessments to evaluate conceptual and practical understanding, and select modules required submission of project documentation and assignments. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

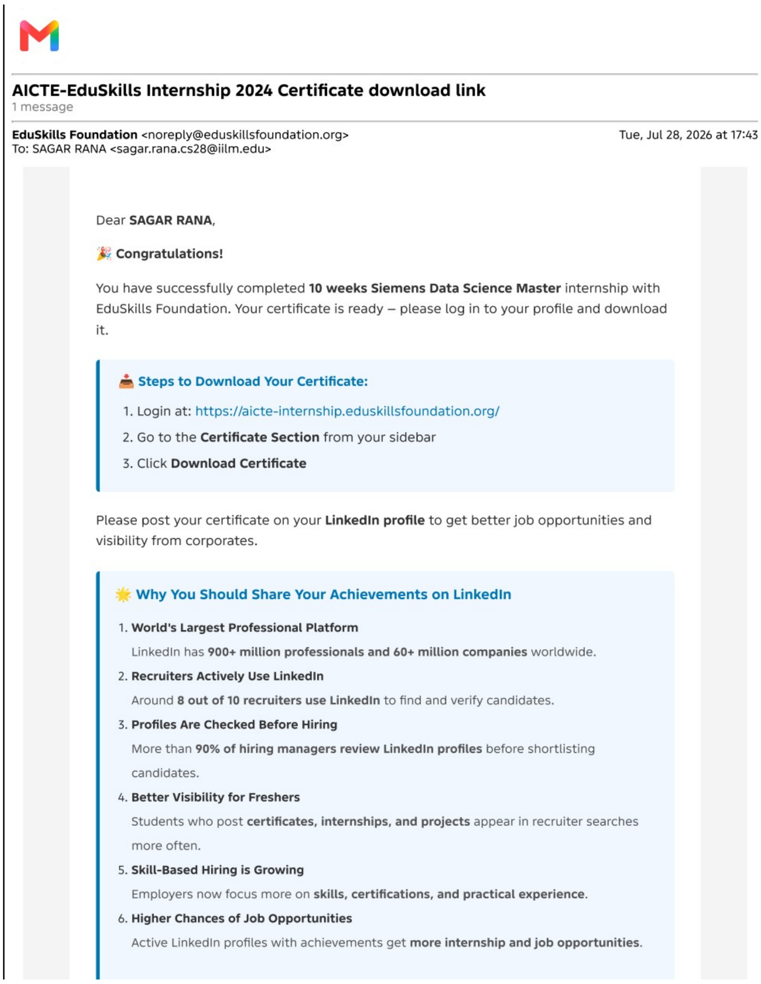
4.9 Expected Outcomes

- Ability to identify business problems suitable for AI and machine learning solutions.
- Practical understanding of data engineering – accessing, loading, transforming, and calculating data.
- Capability to build, evaluate, and improve foundational and advanced machine learning models.

- Hands-on exposure to deploying and monitoring machine learning models.
- Skill to configure and manage an enterprise AI platform (RapidMiner AI Hub) and perform platform administration operations.
- Successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Outstanding” grade.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Siemens Data Science Master, are attached in Chapter 3 of this report. The official completion e-mail from EduSkills Foundation, confirming the certificate is ready for download, is attached below as further communication proof.



AICTE-EduSkills Internship 2024 Certificate download link
1 message

EduSkills Foundation <noreply@eduskillsfoundation.org> Tue, Jul 28, 2026 at 17:43
To: SAGAR RANA <sagar.rana.cs28@iilm.edu>

Dear **SAGAR RANA**,

🎉 **Congratulations!**

You have successfully completed **10 weeks Siemens Data Science Master** internship with EduSkills Foundation. Your certificate is ready – please log in to your profile and download it.

📥 **Steps to Download Your Certificate:**

1. Login at: https://aicte-internship.eduskillsfoundation.org/
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

🌟 **Why You Should Share Your Achievements on LinkedIn**

1. **World's Largest Professional Platform**
   LinkedIn has **900+ million professionals and 60+ million companies** worldwide.
2. **Recruiters Actively Use LinkedIn**
   Around **8 out of 10 recruiters use LinkedIn** to find and verify candidates.
3. **Profiles Are Checked Before Hiring**
   More than **90% of hiring managers review LinkedIn profiles** before shortlisting candidates.
4. **Better Visibility for Freshers**
   Students who post **certificates, internships, and projects** appear in recruiter searches more often.
5. **Skill-Based Hiring is Growing**
   Employers now focus more on **skills, certifications, and practical experience.**
6. **Higher Chances of Job Opportunities**
   Active LinkedIn profiles with achievements get **more internship and job opportunities.**

5. BIBLIOGRAPHY/REFERENCES

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- RapidMiner Documentation, https://docs.rapidminer.com/
- AICTE National Internship Portal, https://internship.aicte-india.org/
- EduSkills Foundation, https://eduskillsfoundation.org/